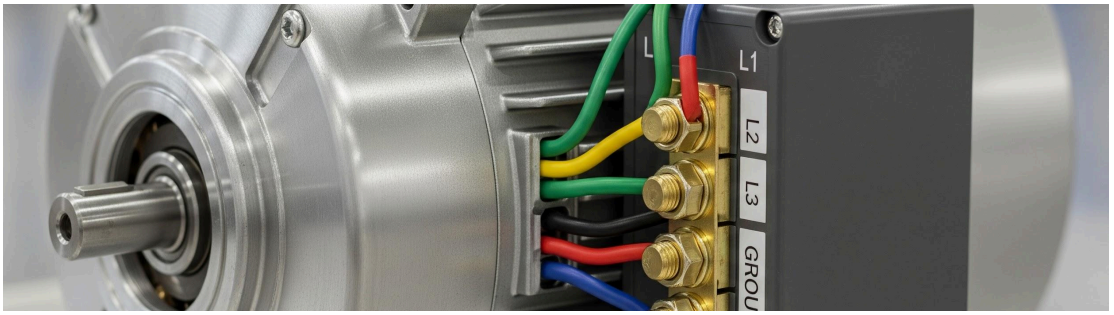

Electrical Machine Data Sheet

Machine Name/Type : DC Shunt Motor Coupled with Alternator

Name plate Details

The rating section details the machine's operational capacity and design limits.

PARAMETER	DC SHUNT MOTOR	ALTERNATOR
Power output	5 HP	3 KVA
Voltage	220 V	415 V
Rated Current	19 A	4.2 A
Field Current	1 A	1.4 A
Speed	1500 RPM	1500 RPM
Frequency	-	50 Hz
Phase	-	3

Extn. Volt	-	220 V
Power Factor	-	0.8

Precautions

I. Ensure **220 V DC supply** is applied to the DC motor before starting.

II. Never run the DC motor **without shunt field excitation**.

III. Use a proper **starter** to limit starting current of the DC motor.

IV. Ensure proper mechanical coupling and shaft alignment.

V. Do not connect electrical load to alternator before it reaches rated speed.

VI. Maintain correct excitation of alternator field winding.

VII. Ensure proper earthing of both motor and alternator.

VIII. Do not exceed rated current of motor or alternator.

IX. Monitor voltage, current, speed, and temperature continuously.

X. Switch OFF immediately in case of abnormal noise, vibration, or overheating.

Working Principle of DC Shunt Motor Coupled with Alternator

The DC shunt motor operates on the principle that a **current-carrying conductor placed in a magnetic field experiences a force**, producing mechanical rotation.

The motor shaft is mechanically coupled to the alternator shaft. As the DC motor rotates, it drives the alternator at synchronous speed.

The alternator works on the principle of **electromagnetic induction**, where a rotating magnetic field induces alternating voltage in the stator windings.
Thus, **DC electrical energy is converted into mechanical energy and then into AC electrical energy**.

Working Characteristics

I. Motor maintains nearly constant speed under varying load.

II. Alternator output voltage increases with field excitation.

III. Frequency of generated AC depends on speed of rotation.

IV. Smooth and stable operation under normal conditions.

V. Overall efficiency depends on losses in both machines.

Applications

1. Laboratory demonstration of AC generation
2. Study of alternator characteristics
3. Speed–frequency relationship experiments
4. Educational training setups
5. Load testing of alternators
6. Research and testing laboratories
7. Power system training modules
8. Generator excitation studies
9. Electrical machine calibration
10. Teaching electromechanical energy conversion